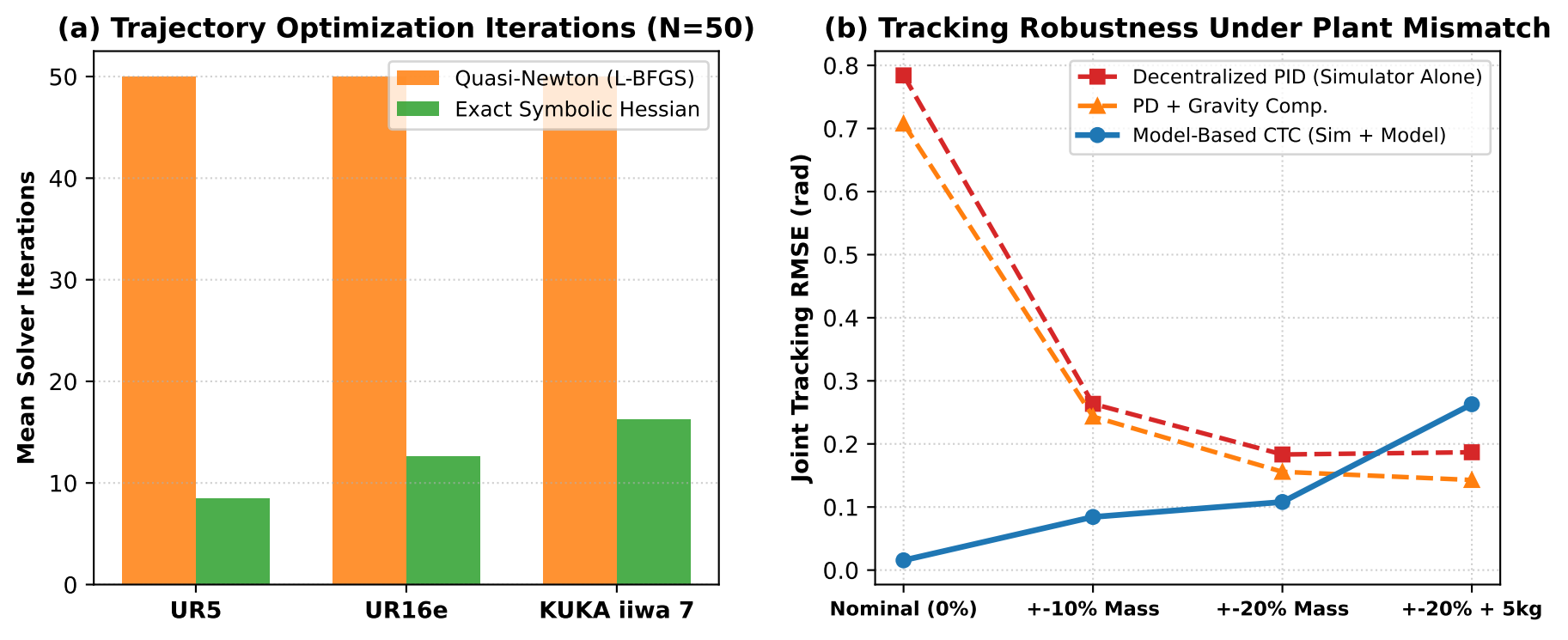
(a) Trajectory Optimization Iterations (N=50)
Mean Solver Iterations
0
10
20
30
40
50
Quasi-Newton (L-BFGS)
Exact Symbolic Hessian
UR5
UR16e
KUKA iiwa 7
(b) Tracking Robustness Under Plant Mismatch
Joint Tracking RMSE (rad)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Decentralized PID (Simulator Alone)
PD + Gravity Comp.
Model-Based CTC (Sim + Model)
Nominal (0%)
+-10% Mass
+-20% Mass
+-20% + 5kg